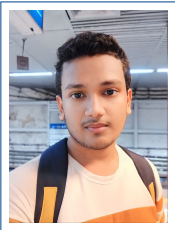


Akshar Nath Gorain

Systems Engineer & AI Researcher

Email: workforakng@gmail.com Phone: +91 90656 01445
LinkedIn: linkedin.com/in/ak-n-g GitHub: github.com/workforakng
Portfolio: workforakng.github.io Location: Kolkata, India



PROFESSIONAL SUMMARY

B.Tech CSE (2024–2028, 3rd Year) bridging AI research with production engineering. Focus on Computer Vision, efficient deep learning, and cloud-connected full-stack systems. Co-author on edge AI and medical imaging research, presented at IEEE SPACE 2026 and IndiSetu 2026. International champion at COMSYS Hackathon 7.

TECHNICAL SKILLS

Languages	Python, JavaScript/TypeScript, C/C++, Dart, Solidity, SQL, Bash/Shell, PHP
AI / CV	PyTorch, YOLOv8, ConvNeXt, EfficientNet, Swin Transformer, SLIC Superpixels, SAM family (MedSAM, MobileSAM), Gemini API
Cloud & Data	AWS (Data Engineering Certified), Google Cloud (Vision, Functions), Firebase, PostgreSQL (Neon), MySQL, Prisma, ETL Pipelines
Web & Mobile	Next.js 14, FastAPI, Flask, Express.js, Node.js, React Native, Expo, Flutter
Tools	Git, Docker, Linux/Termux Automation, OpenMMLab

HONORS & AWARDS

COMSYS Hackathon 7 – International Championship Champion – Rank #1	Škoda Auto University, Czech Republic June 2026
---	--

- Announced at the 7th International Conference on Frontiers in Computing and Systems (COMSYS 2026).
- Hybrid vision pipeline (YOLOv8, MSER, ConvNeXt/EfficientNet ensemble) for multiscript recognition.
- HexaFalls 2, Accio Relevance, JIS University – 3rd Prize (USD 200), July 2026.
- Made in JIS, Celeb Edition 2026, Kolkata – Team CURAA, Aug 2026.
- Tech Darshi – Edge-AI research coverage, Sept 2026.

RESEARCH & PUBLICATIONS

Progressive Knowledge Distillation for Military Vehicle Classification on Edge Devices Techno India University, with Jadavpur University	Co-author 2026
---	-------------------

- CNN-to-CNN, CNN-to-Transformer and multi-teacher distillation with INT8 quantization and mobile deployment. Presented at **IEEE SPACE 2026, Bangalore**.

Liver Tumor Segmentation from CT Scans using SLIC Superpixel based Swin-UNet Techno India University	Co-author 2026
---	-------------------

- SLIC consensus post-processing with fine-tuned Swin-UNet, compared against MedSAM, SAM2-Tiny, MobileSAM and PicoSAM on LiTS. Poster at IndiSetu 2026 AI in Healthcare Summit, Kolkata.

Multiscript Handwritten Character Segmentation & Recognition COMSYS 2026, Škoda Auto Univ. / Jadavpur Univ.	Lead Author 2026
--	---------------------

- Lazy Dataset training, YOLOv8-MSER triple-layer detection, and ConvNeXt-EfficientNet ensemble with script-aware filtering. International Rank #1 (Score: 0.755).

PROFESSIONAL EXPERIENCE

Techno India University Undergraduate AI Researcher	Kolkata, India Aug 2024 – Present
--	--------------------------------------

- Efficient deep learning for low-compute deployment and medical imaging pipelines.

CURAA Mobile App & ML Engineer	Kolkata, India Nov 2025 – Present
-----------------------------------	--------------------------------------

- Engineered cross-platform Flutter architecture with native 60fps hardware rendering and intelligent pet-care service pipelines.

AICTE - IBM SkillsBuild

AI Automation & Intelligent Solutions Intern

Virtual

June 2026 – July 2026

- Worked on automation workflows, AI agents and enterprise integrations.

AICTE - EduSkills (Palo Alto)

Cybersecurity Intern

Virtual

June 2026 – Aug 2026

- Coursework in network security, threat intelligence and security automation.

AICTE - EduSkills Foundations

Virtual Research Intern Stack

Virtual

Dec 2025 – June 2026

- Ethical Hacking, Generative AI with multi-modal pipelines on Google Cloud, and Data Engineering with ETL on AWS.

AICTE Internship

Government Machine Learning Intern

Remote

Past

- ML models for six major water pollutants from historical monitoring data, with evaluation and inference workflow.

PROJECTS

ShareBite Platform

Full-Stack Lead Developer

Next.js 14, PostgreSQL, NextAuth.js

sharebite-platform.vercel.app

- Production food rescue engine aligned with UN SDG 2 with real-time surplus ledger and automated category safety windows.

Delta Terminal

Systems & Quant Architect

Python, Flask, AsyncIO, Socket.IO

- Institutional algo-trading system with Δ -THETA options harvesting, Five-Gate regime filter (ADX, IV Rank, Vol Ratio), and multi-alpha interference engine on Linux/ARM64.

CURAA Mobile App

Lead Mobile Engineer

Flutter, Dart, Node.js, MongoDB

curaa.co.in

- Unified digital ecosystem orchestrating 5 pet-care microservices, veterinary scheduling, and native 60fps dynamic UI flows.

Water Quality Prediction

ML Developer

Python, Machine Learning

- Multi-target regression models predicting six major water pollutants from sensor station data, built for environmental forecasting.

SSH Termux Automation

Automation Developer

Bash, Linux, Termux

github.com/workforakng/ssh_termux_auto

- Automated server telemetry and intrusion detection daemon with dynamic IP tracking and instant Telegram bot alerts.

CERTIFICATIONS

- **AWS Academy Graduate - Data Engineering** (Dec 2025) – AWS Training and Certification.
- **Google Cloud Skill Badges:** Develop GenAI Apps with Gemini & Analyze Images with Cloud Vision API.
- **AI and Data Analytics focused on Green Skills** – Completion certificate.
- **AICTE IBM SkillsBuild** – AI Automation and Intelligent Solutions (June–July 2026).
- **AICTE EduSkills Palo Alto** – Cybersecurity (June–Aug 2026).
- **AICTE EduSkills** – Data Engineering, Generative AI, Ethical Hacking.

EDUCATION

Techno India University

B.Tech - Computer Science and Engineering

Kolkata, WB

2024 – 2028

- **Sem 1:** SGPA 7.67 – **Sem 2:** SGPA 8.49, CGPA 8.08 – **Sem 3:** SGPA 7.37, CGPA 7.84.
- **Coursework:** Machine Learning, Deep Learning, Data Structures, Algorithms, Cloud Computing.

Guru Gobind Singh Public School

CBSE Secondary (2021) & Higher Secondary (2023)

Bokaro, JH